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Voice processing system, voice processing method, and recording medium having voice processing program recorded thereon

Abstract

A voice processing system according to an embodiment includes: an estimation processing unit which executes transmission and reception of voice data between microphone-speaker devices and a communication device, thereby estimating the position of each of the microphone-speaker devices relative to the communication device; and a display processing unit which displays, on a display, the position of the microphone-speaker device estimated by the estimation processing unit.

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2022-109688 filed on Jul. 7, 2022, the entire contents of which are incorporated herein by reference.

BACKGROUND

(2) The present disclosure relates to a voice processing system which performs transmission and reception of voice by a microphone-speaker device, a voice processing method, and a recording medium having a voice processing program recorded thereon.

(3) Conventionally, neck-hanging-type microphone-speaker devices which can be worn around the user's neck are known. By using the microphone-speaker device as mentioned above, the user can hear reproduced voice without having his or her own ears being plugged and can also have the speech voice collected without preparing a device for sound collection.

(4) The microphone-speaker device is connected to a control device such as a personal computer, for example, and the control device can display setting information of the microphone-speaker device. For example, the control device displays identification information (a device number), a microphone gain, and a speaker volume of the microphone-speaker device.

(5) Here, for example, a case where a plurality of users respectively wear the microphone-speaker devices and conduct a meeting is assumed. When a plurality of microphone-speaker devices are used simultaneously within the same area as in the above case, if the device numbers of the respective microphone-speaker devices are displayed on the control device, it becomes difficult for the users to ascertain the microphone-speaker device that the user himself/herself is wearing, and it is also difficult to ascertain the setting content.

SUMMARY

(6) An object of the present disclosure is to provide a voice processing system which allows, when microphone-speaker devices are used, a user to easily ascertain each microphone-speaker device, a voice processing method, and a recording medium having a voice processing program recorded thereon.
(7) A voice processing system according to one aspect of the present disclosure pertains to a system including a microphone-speaker device of a portable type carried by a user, and a communication device installed in a predetermined area in which the user is accommodated. The voice processing system is provided with an estimation processing unit and a display processing unit. The estimation processing unit estimates the position of the microphone-speaker device relative to the communication device by executing transmission and reception of data between the microphone-speaker device and the communication device. The display processing unit displays, on a display, the position of the microphone-speaker device estimated by the estimation processing unit.
(8) A voice processing method according to another aspect of the present disclosure pertains to a method which involves use of a microphone-speaker device of a portable type carried by a user, and a communication device installed in a predetermined area in which the user is accommodated. In the voice processing method, one or more processors execute: estimating the position of the microphone-speaker device relative to the communication device by performing transmission and reception of data between the microphone-speaker device and the communication device; and displaying the estimated position of the microphone-speaker device on a display.
(9) A recording medium according to yet another aspect of the present disclosure pertains to a recording medium having recorded thereon a voice processing program comprising instructions for a microphone-speaker device of a portable type carried by a user, and a communication device installed in a predetermined area in which the user is accommodated. Further, the voice processing program is a program for causing one or more processors to execute: estimating the position of the microphone-speaker device relative to the communication device by performing transmission and reception of data between the microphone-speaker device and the communication device; and displaying the estimated position of the microphone-speaker device on a display.
(10) According to the present disclosure, it is possible to provide a voice processing system which allows, when microphone-speaker devices are used, a user to easily ascertain each microphone-speaker device, a voice processing method, and a recording medium having a voice processing program recorded thereon.
(11) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description with reference where appropriate to the accompanying drawings. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram illustrating a configuration of a voice processing system according to an embodiment of the present disclosure.
(2) FIG. **2** is a diagram illustrating an example of application of the voice processing system according to the embodiment of the present disclosure.
(3) FIG. **3** is an external view illustrating a configuration of a microphone-speaker device according to the embodiment of the present disclosure.
(4) FIG. **4** is a table showing an example of setting information used by the voice processing system according to the embodiment of the present disclosure.

(5) FIG. **5** is a diagram showing an example of a method of estimating the position of the microphone-speaker device according to the embodiment of the present disclosure.
(6) FIG. **6** is a diagram showing an example of a setting screen displayed in the voice processing system according to the embodiment of the present disclosure.
(7) FIG. **7** is a diagram showing an example of a setting screen displayed in the voice processing system according to the embodiment of the present disclosure.
(8) FIG. **8** is a flowchart for illustrating an example of a procedure of voice control processing executed in the voice processing system according to the embodiment of the present disclosure.
(9) FIG. **9** is a diagram showing an example of a setting screen displayed in a voice processing system according to an embodiment of the present disclosure.
(10) FIG. **10** is a diagram showing an example of a setting screen displayed in a voice processing system according to an embodiment of the present disclosure.
(11) FIG. **11** is a diagram showing an example of a setting screen displayed in a voice processing system according to an embodiment of the present disclosure.
(12) FIG. **12** is a diagram showing an example of a setting screen displayed in a voice processing system according to an embodiment of the present disclosure.
(13) FIG. **13** is a diagram showing an example of a setting screen displayed in a voice processing system according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(14) Embodiments of the present disclosure will be described below with reference to the accompanying drawings. Note that the following embodiments are merely examples that embody the present disclosure, and are not intended to limit the technical scope of the disclosure.
(15) A voice processing system according to the present disclosure can be applied to, for example, a case where a plurality of users conduct a meeting by using microphone-speaker devices, respectively, in a meeting room. The microphone-speaker device is a portable audio device carried by the user. Further, the microphone-speaker device has a neckband shape, for example, and each user wears the microphone-speaker device around his/her neck to attend the meeting. Each user can hear a voice reproduced from a speaker of the microphone-speaker device, and can also cause a microphone of the microphone-speaker device to collect a voice uttered by the user himself/herself. Note that the voice processing system according to the present disclosure can also be applied to a case of conducting an online meeting in which a plurality of users at multiple sites use microphone-speaker devices, respectively, and transmit and receive voice data via a network.
(16) Voice Processing System **100**
(17) FIG. **1** is a diagram illustrating a configuration of a voice processing system **100** according to an embodiment of the present disclosure. The voice processing system **100** includes a control device **1**, a microphone-speaker device **2**, and an external speaker **3**. The microphone-speaker device **2** is an audio device equipped with a microphone **24** and a speaker **25** (see FIG. **3**). The microphone-speaker device **2** may have the function of an AI speaker, a smart speaker, and the like, for example. The voice processing system **100** is a system which includes a plurality of wearable microphone-speaker devices **2** to be worn by a plurality of users, respectively, and executes transmission and reception of voice data corresponding to the user's speech voice among the plurality of microphone-speaker devices **2**. The voice processing system **100** is an example of the voice processing system of the present disclosure.
(18) The control device **1** controls each of the microphone-speaker devices **2** and executes, when a meeting is started in a meeting room, for example, processing of transmitting and receiving voices between each of the microphone-speaker devices **2**. Note that the control device **1** alone may constitute the voice processing system according to the present disclosure. When the voice processing system of the present disclosure is to be constituted by the control device **1** alone, the control device **1** may accumulate voices acquired from the microphone-speaker device **2** as voice for recording, and execute processing of having the acquired voice recognized within the device

itself (i.e., voice recognition processing). Further, the voice processing system of the present disclosure may include various servers which provide various services such as a meeting service, a subtitling service by voice recognition, a translation service, and a minutes service.
(19) The external speaker **3** is installed, for example, in a meeting room, and can reproduce a voice output from the control device **1**, a voice which has been collected by the microphone-speaker device **2**, and the like.
(20) The present embodiment will be described with reference to a meeting indicated in FIG. **2** as an example. Users A, B, C, and D, who are attendees of the meeting mentioned above, wear microphone-speaker devices **2**A, **2**B, **2**C, and **2**D, respectively, around their necks in a meeting room R**1**, and attend the meeting. Further, the control device **1** and external speakers **3***a* and **3***b* are installed in the meeting room R**1**. The control device **1** and the microphone-speaker devices **2**A, **2**B, **2**C, **2**D are connected to each other by wireless communication such as Bluetooth (registered trademark). The control device **1** and the external speakers **3***a* and **3***b* are connected to each other via an audio cable, a USB cable, a wired LAN, etc.
(21) Specifically, when the control device **1** acquires data corresponding to speech voice of user A from the microphone-speaker device **2**A, the control device **1** may transmit the voice data to each of the microphone-speaker devices **2**B, **2**C, and **2**D of users B, C, and D, and cause each of the microphone-speaker devices **2**B, **2**C, and **2**D to reproduce the speech voice. Further, when the control device **1** acquires data corresponding to speech voice of user B from the microphone-speaker device **2**B, the control device **1** may transmit the voice data to each of the microphone-speaker devices **2**A, **2**C, and **2**D of users A, C, and D, and cause each of the microphone-speaker devices **2**A, **2**C, and **2**D to reproduce the speech voice. Furthermore, the control device **1** may cause each of the external speakers **3***a* and **3***b* to reproduce, from the external speakers **3***a* and **3***b*, a speech voice acquired from the microphone-speaker device **2**.
(22) Microphone-Speaker Device **2**
(23) FIG. **3** illustrates an example of the outer appearance of the microphone-speaker device **2**. As illustrated in FIG. **3**, the microphone-speaker device **2** is provided with a power source **22**, a connection button **23**, the microphone **24**, the speaker **25**, and a communicator (not shown). The microphone-speaker device **2** is, for example, a neckband-type wearable device that can be worn around the user's neck. The microphone-speaker device **2** acquires a voice of a user through the microphone **24** and reproduces (outputs) a voice to the user from the speaker **25**. The microphone-speaker device **2** may be provided with a display which displays various kinds of information.
(24) As illustrated in FIG. **3**, a main body **21** of the microphone-speaker device **2** has left and right arms as seen from the user wearing the microphone-speaker device **2**, and is formed to be U-shaped.
(25) The microphone **24** is disposed at a distal end portion of the microphone-speaker device **2** so that the user's speech voice can be easily collected. The microphone **24** is connected to a microphone substrate (not shown) incorporated in the microphone-speaker device **2**.
(26) The speaker **25** includes, as seen from the user wearing the microphone-speaker device **2**, a speaker **25**L disposed on the left arm and a speaker **25**R disposed on the right arm. Each of the speakers **25**L and **25**R is disposed near the center of the arm of the microphone-speaker device **2** such that the user can easily hear a voice reproduced therefrom. The speakers **25**L and **25**R are connected to a speaker substrate (not shown) incorporated in the microphone-speaker device **2**.
(27) The microphone substrate is a transmitter substrate for transmitting the voice data to the control device **1**, and is included in the communicator. Further, the speaker substrate is a receiver substrate for receiving the voice data from the control device **1**, and is included in the communicator.
(28) The communicator is a communication interface for enabling, wirelessly, execution of data communication according to a predetermined communication protocol between the microphone-speaker device **2** and the control device **1**. Specifically, the communicator performs the

communication by establishing connection between the microphone-speaker device **2** and the control device **1** by a Bluetooth method, for example. For example, when the user presses the connection button **23** after setting the power source **22** to ON, the communicator executes pairing processing to connect the microphone-speaker device **2** to the control device **1**. A transmitter device may be arranged between the microphone-speaker device **2** and the control device **1**. In such a case, the transmitter device may be paired (Bluetooth connected) with the microphone-speaker device **2**, and the transmitter device and the control device **1** may be connected to each other via the Internet.
(29) Control Device **1**
(30) As illustrated in FIG. **1**, the control device **1** is an information processor (e.g., a personal computer) which is provided with a controller **11**, a storage **12**, an operation display **13**, a communicator **14**, and the like. Note that the control device **1** is not limited to being a single computer, but may be a computer system in which a plurality of computers operate in cooperation with each other. Further, various kinds of processing to be executed by the control device **1** may be executed decentrally by one or more processors.
(31) The communicator **14** is a communicator for connecting the control device **1** to a communication network in a wired or wireless manner, thereby executing data communication according to a predetermined communication protocol with the microphone-speaker devices **2** and external devices such as the external speakers **3***a* and **3***b*, via the communication network. For example, the communicator **14** executes the pairing processing by the Bluetooth method, and is connected to the microphone-speaker device **2**. Also, the communicator **14** is connected to the external speakers **3***a* and **3***b* via the audio cable, USB cable, wired LAN, etc.
(32) The operation display **13** is a user interface including a display such as a liquid crystal display or an organic EL display which displays various kinds of information, and an operation portion such as a mouse, a keyboard, or a touch panel which receives operations. The display may be configured separately from the control device **1**, and may be connected to the control device **1** in a wired or wireless manner.
(33) The storage **12** is a non-volatile storage, such as a hard disk drive (HDD) or a solid state drive (SSD), which stores various kinds of information. Specifically, the storage **12** stores data such as setting information D**1** of the microphone-speaker devices **2**.
(34) FIG. **4** illustrates an example of the setting information D**1**. As illustrated in FIG. **4**, the setting information D**1** includes information such as “Device ID”, “Positional Information”, “Volume”, and “Microphone Gain”. The “Device ID” is identification information of each of the microphone-speaker devices **2**. For example, a device number is registered as the device ID. In this example, “MS001” to “MS004” correspond to the microphone-speaker devices **2**A to **2**D, respectively. The “Positional Information” is information which indicates the position of the microphone-speaker device **2** in a meeting room. The controller **11** estimates the respective positions of the microphone-speaker devices **2**A to **2**D in the meeting room and registers the estimated positional information in the setting information D**1**. A method of estimating the position of the microphone-speaker device **2** will be described later.
(35) The “Volume” indicates a reproduction volume (a speaker volume) of the speaker **25** of each of the microphone-speaker devices **2**, and the “Microphone Gain” indicates a gain of the microphone **24** of each of the microphone-speaker devices **2**. The speaker volume and the microphone gain mentioned above are one example of the setting information of the microphone-speaker device of the present disclosure. Note that the setting information may include, in addition to the speaker volume and the microphone gain, information related to functions such as mute, voice recognition, translated voice, and equalizer. The controller **11** registers each piece of the above information in the setting information D**1** every time the microphone-speaker device **2** is connected.
(36) The user can, for example, operate (i.e., operate by a touch) a setting screen F**1** (FIG. **6**, etc.) displayed on the operation display **13** to perform an operation of setting or changing various setting

items (speaker volume, microphone gain, mute, voice recognition, translated voice, equalizer, etc.). The controller **11** registers and updates the setting information D**1** in response to the user operation. In addition, the controller **11** registers the setting information in advance on the basis of the user operation or default settings.
(37) Further, the storage **12** stores a control program such as a voice control program for causing the controller **11** to execute voice control processing (FIG. **8**) to be described later. For example, the voice control program may be recorded, in a non-transitory way, on a computer-readable recording medium such as a CD or a DVD, and be read by a reading device (not shown) such as a CD drive or a DVD drive provided in the control device **1** to be stored in the storage **12**.
(38) The controller **11** includes control devices such as a CPU, a ROM, and a RAM. The CPU is a processor which executes various kinds of arithmetic processing. The ROM is a non-volatile storage which stores in advance control programs such as a BIOS and an OS for causing the CPU to execute the various kinds of arithmetic processing. The RAM is a volatile or non-volatile storage which stores various kinds of information, and is used as a temporary storage memory (a work area) for various kinds of processing to be executed by the CPU. Further, the controller **11** controls the control device **1** by using the CPU to execute various control programs stored in advance in the ROM or the storage **12**.
(39) Specifically, the controller **11** includes, as illustrated in FIG. **1**, various processing units such as an estimation processing unit **111**, a display processing unit **112**, a reception processing unit **113**, and a setting processing unit **114**. The controller **11** functions as the above various processing units by using the CPU to execute various kinds of processing according to the control program. Moreover, some or all of the processing units may be configured by an electronic circuit. Note that the control program may be a program for causing a plurality of processors to function as the above-mentioned processing units.
(40) The estimation processing unit **111** estimates the position of each of the plurality of microphone-speaker devices **2**. Specifically, the estimation processing unit **111** estimates the position of each of the microphone-speaker devices **2** in a predetermined area (in this case, the meeting room R**1**) where the plurality of microphone-speaker devices **2** and the external speakers **3***a* and **3***b* are arranged. For example, the estimation processing unit **111** estimates the position of each of the microphone-speaker devices **2**A to **2**D relative to the external speakers **3***a* and **3***b* in the meeting room R**1**.
(41) FIG. **5** illustrates an example of a method of estimating the position of the microphone-speaker device **2**. For example, as illustrated in FIG. **5**, the estimation processing unit **111** makes a first specific sound (e.g., a test sound) be reproduced from the external speaker **3***a*. The first specific sound is input to (collected through) each of the microphone **24** of the microphone-speaker device **2**A, the microphone **24** of the microphone-speaker device **2**B, the microphone **24** of the microphone-speaker device **2**C, and the microphone **24** of the microphone-speaker device **2**D. The estimation processing unit **111** acquires, from each of the microphone-speaker devices **2**A to **2**D, the first specific sound that has been collected. The estimation processing unit **111** measures the length of time from when the external speaker **3***a* is made to reproduce the first specific sound to when the first specific sound is acquired by each of the microphone-speaker devices **2**A to **2**D, thereby calculating a distance from the external speaker **3***a* to each of the microphone-speaker devices **2**A to **2**D.
(42) Similarly, the estimation processing unit **111** makes a second specific sound be reproduced from the external speaker **3***b*. The second specific sound is input to (collected through) each of the microphone **24** of the microphone-speaker device **2**A, the microphone **24** of the microphone-speaker device **2**B, the microphone **24** of the microphone-speaker device **2**C, and the microphone **24** of the microphone-speaker device **2**D. The estimation processing unit **111** acquires, from each of the microphone-speaker devices **2**A to **2**D, the second specific sound that has been collected. The estimation processing unit **111** measures the length of time from when the external speaker **3***b* is

made to reproduce the second specific sound to when the second specific sound is acquired by each of the microphone-speaker devices **2**A to **2**D, thereby calculating a distance from the external speaker **3***b* to each of the microphone-speaker devices **2**A to **2**D.

(43) Further, the estimation processing unit **111** estimates the position of each of the microphone-speaker devices **2**A to **2**D relative to the external speakers **3***a* and **3***b*, on the basis of the distance from the external speaker **3***a* to each of the microphone-speaker devices **2**A to **2**D and the distance from the external speaker **3***b* to each of the microphone-speaker devices **2**A to **2**D.

(44) As can be seen, the estimation processing unit **111** estimates the position of each of the plurality of microphone-speaker devices **2** relative to the external speaker **3** by executing transmission and reception of voice data between each of the plurality of microphone-speaker devices **2** and the external speaker **3**. The estimation processing unit **111** registers the positional information indicating the estimated position of each of the microphone-speaker devices **2**A to **2**D in the setting information D**1** (FIG. **4**). The method of estimating the position of the microphone-speaker device **2** is not limited to the above-described method, but may be any of the other methods to be described later. Alternatively, the estimation processing unit **111** may execute transmission and reception of voice data between a single microphone-speaker device **2** and the external speaker **3**, thereby estimating the position of the microphone-speaker device **2** relative to the external speaker **3**.

(45) The display processing unit **112** displays, on the operation display **13**, the respective positions of the plurality of microphone-speaker devices **2** estimated by the estimation processing unit **111**. Also, the display processing unit **112** displays, on the operation display **13**, the setting screen F**1** for setting each of setting items of the plurality of microphone-speaker devices **2**. In addition, the display processing unit **112** displays, on the operation display **13**, the setting information of each of the plurality of microphone-speaker devices **2**.

(46) FIG. **6** illustrates an example of the setting screen F**1**. For example, as illustrated in FIG. **6**, the display processing unit **112** displays, on the setting screen F**1**, images corresponding to the external speakers **3***a* and **3***b* installed in the meeting room R**1** and images corresponding to the microphone-speaker devices **2**A to **2**D, respectively. Specifically, the display processing unit **112** displays, on the setting screen F**1**, the respective images of the microphone-speaker devices **2**A to **2**D at display positions corresponding to the positions estimated by the estimation processing unit **111**. The display processing unit **112** may display identification information (the device number or the like) (see FIG. **4**) of the microphone-speaker device **2** for each of the images of the microphone-speaker devices **2**. Also, if user information (user ID, user name, etc.) is registered for each microphone-speaker device **2**, the display processing unit **112** may display the user information at each of the images of the microphone-speaker devices **2**.

(47) According to the setting screen F**1** illustrated in FIG. **6**, the user can easily ascertain the position of each of the microphone-speaker devices **2** with respect to the external speakers **3***a* and **3***b*. Note that the display processing unit **112** may display, on the operation display **13**, the position of a single microphone-speaker device **2** estimated by the estimation processing unit **111**.

(48) The reception processing unit **113** receives, on the operation display **13**, a setting operation for the setting items of each of the plurality of microphone-speaker devices **2** from the user. For example, on the setting screen F**1** illustrated in FIG. **6**, the reception processing unit **113** receives the setting operation for the setting items of the microphone-speaker device **2** from the user. Specifically, when the user selects (by a touch operation, a click operation, etc.) the microphone-speaker device **2**A on the setting screen F**1** illustrated in FIG. **6**, for example, the reception processing unit **113** displays a plurality of setting items of the microphone-speaker device **2**A. In this case, the reception processing unit **113** displays the setting items, which are “Mute”, “Voice Recognition”, “Reproduction of Translated Voice”, and “Language of Translated Voice”, and receives the setting operation from the user. FIG. **6** illustrates the state in which the user has made the setting for “Mute” of the microphone-speaker device **2**A to “OFF”.

(49) The setting processing unit **114** sets the setting items of each of the microphone-speaker devices **2** according to the setting operation by the user. For example, when the user selects the microphone-speaker device **2**A and performs the setting operation for each of the setting items, the setting processing unit **114** sets each of the setting items for the microphone-speaker device **2**A. Further, for example, when the user selects the microphone-speaker device **2**B and performs the setting operation for each of the setting items, the setting processing unit **114** sets each of the setting items for the microphone-speaker device **2**B. In this way, the setting processing unit **114** sets each of the setting items on the basis of the setting operation by the user for each of the microphone-speaker devices **2** on the setting screen F**1**.
(50) In the example illustrated in FIG. **2**, an organizer of the meeting, for example, collectively performs the setting operations for the respective microphone-speaker devices **2**A to **2**D, and the setting processing unit **114** sets each of the microphone-speaker devices **2**A to **2**D on the basis of the setting operations. In addition, the setting processing unit **114** may set the speaker volume and the microphone gain, etc., according to the setting operation by the user for each of the microphone-speaker devices **2**A to **2**D.
(51) The setting processing unit **114** registers the setting content of each of the microphone-speaker devices **2**A to **2**D in the setting information D**1** (FIG. **4**).
(52) When the setting processing unit **114** completes the setting for each of the microphone-speaker devices **2**A to **2**D, users A to D, for example, can conduct a meeting using the microphone-speaker devices **2**A to **2**D. For example, when user A speaks, the controller **11** may acquire the speech voice of user A from the microphone-speaker device **2**A and make that speech voice be reproduced from each of the microphone-speaker devices **2**B to **2**D. Further, for example, when user B speaks, the controller **11** may acquire the speech voice of user B from the microphone-speaker device **2**B and make that speech voice be reproduced from each of the microphone-speaker devices **2**A, **2**C, and **2**D. For example, in a situation in which users B to D are wearing earphones or the like and it is difficult to hear the speech voice of user A, the controller **11** may make the speech voice of user A acquired from the microphone-speaker device **2**A be reproduced from each of microphone-speaker devices **2**B to **2**D. Further, for example, if a language used by user A is different from languages used by users B to D, the controller **11** may make a voice corresponding to a translation of the speech voice of user A be reproduced from each of the microphone-speaker devices **2**B to **2**D. In this way, the controller **11** may transmit and receive voice data between the microphone-speaker devices **2**A to **2**D on the basis of the setting content registered in the setting information D**1**. In addition, the controller **11** may make the voice data be reproduced from the external speakers **3***a* and **3***b*, if necessary. Further, the controller **11** may, for example, make the speech voice of a user at a different site (a remote location) from the meeting room R**1** be reproduced from the microphone-speaker devices **2**A to **2**D.
(53) Here, the setting processing unit **114** may change the setting content of the microphone-speaker devices **2**A to **2**D during the meeting. Specifically, the setting processing unit **114** sets a specific place relative to the external speakers **3***a* and **3***b* to a specific area AR**1**. For example, the setting processing unit **114** sets an area including the microphone-speaker devices **2**A to **2**D in the meeting room R**1** (FIG. **2**) as the specific area AR**1**. The display processing unit **112** displays the specific area AR**1** on the setting screen F**1** (FIG. **7**). In a state in which the specific area AR**1** is set, when the microphone-speaker device **2**D is moved outside the specific area AR**1** from within the specific area AR**1** (see FIG. **7**) as user D who is wearing the microphone-speaker device **2**D leaves the meeting room R**1** during the meeting, for example, the setting processing unit **114** changes the setting content of the microphone-speaker device **2**D. For example, the setting processing unit **114** changes the setting content for mute of the microphone-speaker device **2**D from “OFF” to “ON”. In other words, the setting processing unit **114** sets the microphone **24** and the speaker **25** of the microphone-speaker device **2**D to OFF.
(54) As a result, voice is no longer reproduced from the speaker **25** of the microphone-speaker

device **2**D, and the speech voice of user D is also no longer collected by the microphone **24**. This allows, for example, user D to leave his/her seat with the microphone-speaker device **2**D being worn even when user D needs to leave his/her seat to answer a phone call during the meeting, and there is also no need for user D to perform an operation to change the setting content of the microphone-speaker device **2**D himself/herself.

(55) As described above, when the setting processing unit **114** has set the specific area AR**1**, the setting processing unit **114** may vary the setting content between the microphone-speaker device **2** located within the specific area AR**1** among the plurality of microphone-speaker devices **2** and the microphone-speaker device **2** located outside the specific area AR**1** among the plurality of microphone-speaker devices **2**. Note that the setting processing unit **114** may set the setting items of a single microphone-speaker device **2** according to the setting operation by the user.

(56) Further, when the microphone-speaker device **2**D is moved outside the specific area AR**1** from within the specific area AR**1**, the display processing unit **112** displays, on the setting screen F**1**, an image of the microphone-speaker device **2**D outside the specific area AR**1** (see FIG. **7**). This allows the user to keep track of changes in the position of each of the microphone-speaker devices **2**. When the user selects the microphone-speaker device **2**D on the setting screen F**1** illustrated in FIG. **7**, the display processing unit **112** displays the changed setting content (Mute: ON).

(57) Voice Control Processing

(58) Now, with reference to FIG. **8**, an example of a procedure of voice control processing executed by the controller **11** of the control device **1** will be described.

(59) The present disclosure can be regarded as a voice control method of executing one or more steps included in the voice control processing (i.e., a voice processing method of the present disclosure). Further, the one or more steps included in the voice control processing described herein may be omitted as appropriate. Furthermore, the order in which the steps of the voice control processing are executed may be varied as long as the same effect and advantage are produced. Moreover, although a case where the controller **11** executes each step of the voice control processing will be described herein as an example, in other embodiments, one or more processors may execute each step of the voice control processing in a decentralized manner.

(60) First, in step S**1**, the controller **11** estimates the position of each of the microphone-speaker devices **2**. In the example illustrated in FIG. **2**, when each of the microphone-speaker devices **2**A to **2**D is connected to the control device **1**, the controller **11** executes processing of estimating the position of each of the microphone-speaker devices **2**A to **2**D. Specifically, the controller **11** makes a first specific sound (a test sound) be reproduced from the external speaker **3***a*. Next, the controller **11** acquires, from each of the microphone-speaker devices **2**A to **2**D, the first specific sound collected by each of the microphone-speaker devices **2**A to **2**D. Next, the controller **11** calculates a distance from the external speaker **3***a* to each of the microphone-speaker devices **2**A to **2**D, on the basis of the length of time from when the external speaker **3***a* reproduces the first specific sound to when the first specific sound is acquired from each of the microphone-speaker devices **2**A to **2**D.

(61) Similarly, the controller **11** makes a second specific sound be reproduced from the external speaker **3***b*. Next, the controller **11** acquires, from each of the microphone-speaker devices **2**A to **2**D, the second specific sound collected by each of the microphone-speaker devices **2**A to **2**D. The controller **11** calculates a distance from the external speaker **3***b* to each of the microphone-speaker devices **2**A to **2**D, on the basis of the length of time from when the external speaker **3***b* is made to reproduce the second specific sound to when the second specific sound is acquired from each of the microphone-speaker devices **2**A to **2**D.

(62) Further, the controller **11** estimates the position of each of the microphone-speaker devices **2**A to **2**D relative to the external speakers **3***a* and **3***b*, on the basis of the distance from the external speaker **3***a* to each of the microphone-speaker devices **2**A to **2**D and the distance from the external speaker **3***b* to each of the microphone-speaker devices **2**A to **2**D. In other words, the controller **11** estimates the position of each of the microphone-speaker devices **2**A to **2**D with respect to the

external speakers **3***a* and **3***b*. The controller **11** registers positional information indicating the estimated position of each of the microphone-speaker devices **2**A to **2**D in the setting information D**1** (FIG. **4**).

(63) Next, in steps S**2**, the controller **11** displays the estimated position of each of the plurality of microphone-speaker devices **2** on the operation display **13**. For example, the controller **11** displays, on the setting screen F**1** illustrated in FIG. **6**, the position of each of the microphone-speaker devices **2**A to **2**D relative to the external speakers **3***a* and **3***b* in an identifiable manner.

(64) Next, in step S**3**, the controller **11** determines whether a selection operation for the microphone-speaker device **2** has been received. For example, on the setting screen F**1** illustrated in FIG. **6**, the controller **11** displays images corresponding to the microphone-speaker devices **2**A to **2**D, respectively, and receives an operation of selecting the microphone-speaker devices **2**A to **2**D from the user. If the controller **11** receives an operation of selecting one of the microphone-speaker devices **2** (S**3**: Yes), the controller **11** shifts the processing to step S**4**. Meanwhile, if the controller **11** does not receive any operation of selecting the microphone-speaker device **2** (S**3**: No), the controller **11** shifts the processing to step S**7**.

(65) In step S**4**, the controller **11** displays setting information of the selected microphone-speaker device **2**. For example, as illustrated in FIG. **6**, when the microphone-speaker device **2**A is selected by the user, the controller **11** displays the current setting information of the microphone-speaker device **2**A on the setting screen F**1**.

(66) Next, in step S**5**, the controller **11** determines whether a setting operation to set the setting items of the microphone-speaker device **2** has been received from the user. For example, as illustrated in FIG. **6**, when the user performs a setting operation to set the setting items of the microphone-speaker device **2**A, the controller **11** receives the setting operation. If the controller **11** receives the setting operation from the user (S**5**: Yes), the controller **11** shifts the processing to step S**6**. Meanwhile, if the controller **11** does not receive the setting operation from the user (S**5**: No), the controller **11** shifts the processing to step S**7**.

(67) In step S**6**, the controller **11** registers the setting content of the microphone-speaker device **2**. For example, as illustrated in FIG. **6**, when the user performs a setting operation to set the setting item for mute of the microphone-speaker device **2**A to “OFF”, the controller **11** sets the setting item for mute of the microphone-speaker device **2**A to “OFF”. The controller **11** registers the setting content in the setting information D**1** (FIG. **4**).

(68) Next, in step S**7**, the controller **11** determines whether the setting operation for the setting items of each of the microphone-speaker devices **2** by the user has been finished. For example, when the user performs a setting operation for the setting items by selecting each of the microphone-speaker devices **2**A to **2**D in order, and then selects a “Done” button (FIG. **6**), the controller **11** determines that the setting operation has been finished (S**7**: Yes) and shifts the processing to step S**8**. When the user does not select the “Done” button, the controller **11** determines that the setting operation has not been finished (S**7**: No) and shifts the processing to step S**3**. The controller **11** repeatedly executes the processing of steps S**3** to S**6** until the setting operation by the user is finished.

(69) Next, in step S**8**, the controller **11** sets a specific area. Specifically, the controller **11** sets a specific place relative to the external speakers **3***a* and **3***b* in the meeting room R**1** to the specific area AR**1**. For example, the controller **11** sets an area including the microphone-speaker devices **2**A to **2**D in the meeting room R**1** (FIG. **2**) as the specific area AR**1**. In this example, the specific area AR**1** may be the meeting room R**1**.

(70) Next, in step S**9**, the controller **11** starts processing of transmitting and receiving voice data between each of the microphone-speaker devices **2**. For example, when the user selects the “Done” button, the controller **11** sets the specific area AR**1** and starts transmission and reception of voice data between each of the microphone-speaker devices **2**A to **2**D. This allows users A to D to conduct a meeting using the microphone-speaker devices **2**A to **2**D. The controller **11** transmits and

receives voice data between the microphone-speaker devices **2**A to **2**D during the meeting.
(71) Next, in step S**10**, the controller **11** determines whether the microphone-speaker device **2** has moved into or out of the specific area AR**1**. If the controller **11** determines that the microphone-speaker device **2** has moved out of the specific area AR**1** or determines that the microphone-speaker device **2** has moved into the specific area AR**1** (S**10**: Yes), the controller **11** shifts the processing to step S**11**. Meanwhile, if the controller **11** determines that the microphone-speaker device **2** has not moved into or out of the specific area AR**1** (S**10**: No), the controller **11** shifts the processing to step S**12**.
(72) The controller **11** repeatedly executes the processing of estimating the position of each of the microphone-speaker devices **2**A to **2**D at predetermined intervals during the meeting, thereby ascertaining the position of each of the microphone-speaker devices **2**A to **2**D in real time.
(73) In step S**11**, the controller **11** changes the setting content of the microphone-speaker device **2**. For example, as illustrated in FIG. **7**, when user D who is wearing the microphone-speaker device **2**D leaves the meeting room R**1** in order to answer a phone call or the like during the meeting, the controller **11** determines that the microphone-speaker device **2**D has been moved outside the specific area AR**1** from within the specific area AR**1** (S**10**: Yes) and changes the setting content for mute of the microphone-speaker device **2**D from "OFF" to "ON" (S**11**).
(74) Also, when user D who has left the meeting room R**1** returns to the meeting room R**1**, the controller **11** determines that the microphone-speaker device **2**D has been moved from outside the specific area AR**1** to within the specific area AR**1** (S**10**: Yes) and changes the setting content for mute of the microphone-speaker device **2**D from "ON" to "OFF" (S**11**).
(75) In step S**12**, the controller **11** determines whether the meeting is ended. For example, when the meeting is ended and connection with each microphone-speaker device **2** is cut, the controller **11** determines that the meeting has been ended. When the meeting is ended (S**12**: Yes), the controller **11** ends the voice control processing. The controller **11** repeatedly executes the processing of steps S**10** and S**11** until the meeting is ended (S**12**: No). The controller **11** may receive a change operation to change the setting content of the microphone-speaker device **2** from the user on the setting screen F**1** during the meeting. In this case, the controller **11** changes the setting content of the microphone-speaker device **2** on the basis of the change operation by the user during the meeting. The controller **11** executes the voice control processing as described above.
(76) As described above, the voice processing system **100** according to the present embodiment is a system including a plurality of portable microphone-speaker devices **2** of a portable type carried by users, and communication devices (e.g., the external speakers **3***a* and **3***b*) installed in a predetermined area (e.g., the meeting room R**1**) in which the users are accommodated. Further, the voice processing system **100** estimates the position of the microphone-speaker device **2** relative to the communication device by executing transmission and reception of voice data between the microphone-speaker device **2** and the communication device, and displays the estimated position of the microphone-speaker device **2** on a display (the operation display **13**). Furthermore, the voice processing system **100** displays, on the display, the setting information of the microphone-speaker device **2**.
(77) According to the above configuration, the user can easily ascertain the position of each of the microphone-speaker devices **2** by using, for example, the external speakers **3***a* and **3***b* installed in the meeting room R**1** as landmarks. Consequently, since the user can easily ascertain the setting content of each of the microphone-speaker devices **2**, the setting items of each of the microphone-speaker devices **2** can be set appropriately.
Other Embodiments
(78) The present disclosure is not limited to the above-described embodiment. Other embodiments of the present disclosure will be described below.
(79) For example, as illustrated in FIG. **9**, a setting processing unit **114** may set a place where a presentation or the like is to be made to a specific area AR**2**. For example, the setting processing

unit **114** sets a space for one user at a front part of a meeting room R**1** as the specific area AR**2** (i.e., a presenter area). A display processing unit **112** displays the specific area AR**2** to be identifiable on a setting screen F**1**, as illustrated in FIG. **9**. Here, the setting processing unit **114** increases a microphone gain of a microphone-speaker device **2** located within the specific area AR**2** so that the microphone-speaker device **2** of a presenter can reliably collect the presenter's speech voice. In the example illustrated in FIG. **9**, since user A is positioned in the specific area AR**2** as the presenter, the setting processing unit **114** sets the microphone gain of a microphone-speaker device **2**A of user A higher than that of microphone-speaker devices **2**B to **2**D.

(80) Further, when user A finishes the presentation and moves outside the specific area AR**2**, the setting processing unit **114** returns the microphone gain of the microphone-speaker device **2**A to the original setting value.

(81) As another embodiment, as illustrated in FIG. **10**, for example, the setting processing unit **114** may further set, in addition to the specific area AR**2**, a specific area AR**3** at a place near external speakers **3***a* and **3***b*. The display processing unit **112** displays the specific areas AR**2** and AR**3** to be identifiable on the setting screen F**1**, as illustrated in FIG. **10**. For example, the setting processing unit **114** sets a space for one user at the front part of the meeting room R**1** to the specific area AR**2** (the presenter area), and also sets a space at which voices and sounds reproduced from the external speakers **3***a* and **3***b* can be heard to the specific area AR**3**. For example, when a controller **11** makes the presenter's speech voice be reproduced from the external speakers **3***a* and **3***b*, since users near the external speakers **3***a* and **3***b* can hear voice reproduced from the external speakers **3***a* and **3***b*, there is no need to reproduce the voice from a speaker **25** of the microphone-speaker device **2** of these users.

(82) Thus, the setting processing unit **114** increases the microphone gain of the microphone-speaker device **2** located within the specific area AR**2**, and also lowers the speaker volume of the microphone-speaker device **2** located within the specific area AR**3**. In the example illustrated in FIG. **10**, since user A is positioned within the specific area AR**2** as the presenter and user C is positioned within the specific area AR**3** as a listener, the setting processing unit **114** sets the microphone gain of the microphone-speaker device **2**A of user A higher than that of the microphone-speaker devices **2**B to **2**D, and also sets the speaker volume of the microphone-speaker device **2**C of user C lower than that of the microphone-speaker devices **2**A, **2**B, and **2**D. The setting processing unit **114** may set the speaker of the microphone-speaker device **2**C to OFF (“Mute”).

(83) Further, as illustrated in FIG. **11**, the setting processing unit **114** may set a space for one user at the front part of the meeting room R**1** to the specific area AR**2** (the presenter area), and also set a space surrounding a table in the meeting room R**1** to a specific area AR**4**. The specific area AR**4** includes the microphone-speaker devices **2**A to **2**D of users A to D who are seated around the table. A presenter among users A to D moves from the specific area AR**4** to the specific area AR**2**. In this case, the setting processing unit **114** sets the microphone gain of the presenter's microphone-speaker device **2** to high, and also sets the microphone gain of the other users' microphone-speaker devices **2** to low or makes the setting for mute to “ON”.

(84) As described above, the setting processing unit **114** may further set the specific areas AR**3** and AR**4** at specific places relative to the external speakers **3***a* and **3***b*, in addition to the specific area AR**2**, and set each of a plurality of microphone-speaker devices **2** to have the setting content corresponding to the specific area where those microphone-speaker devices **2** are located.

(85) Further, for example, in a case where air-conditioning equipment such as an air conditioner is installed in the meeting room R**1**, the setting processing unit **114** may set a specific area at a place near the air-conditioning equipment. In this case, the setting processing unit **114** may enhance a noise suppressor for a microphone **24** of the microphone-speaker device **2** located within the specific area so as to avoid the influence of noise caused by the air-conditioning equipment.

(86) Furthermore, as yet another embodiment, the setting processing unit **114** may set a plurality of specific areas AR such that the specific areas AR overlap one another. For example, as illustrated in

FIG. **12**, the setting processing unit **114** sets the specific area AR**2** (the presenter area) at the front part of the meeting room R**1**, and also sets a specific area AR**5** at a place including the specific area AR**2**. In this case, the setting processing unit **114** sets the specific area AR**5** to an area with low microphone gain. Here, when user A is positioned within the specific area AR**2** as the presenter, the setting processing unit **114** sets the microphone gain of the microphone-speaker device **2**A to be high. As can be seen, when there exist overlapping specific areas in which the setting contents are different, the setting processing unit **114** may set each of the microphone-speaker devices **2** in accordance with the order of priority corresponding to the use application. In this example, the setting processing unit **114** places a higher priority on the presentation and makes the setting by prioritizing the setting content of the specific area AR**2** over the setting content of the specific area AR**5**.
(87) Further, for example, when users A to D are conducting a meeting in multiple languages, the setting processing unit **114** makes voice of the meeting be reproduced from the microphone-speaker devices **2**B to **2**D located outside the specific area AR**2**, which is in the specific area AR**5**, and makes translated voice corresponding to voice of the meeting be reproduced from the microphone-speaker device **2**A located within the specific area AR**2**, which is in the specific area AR**5**.
(88) In other words, when the specific area AR**2** is included in the specific area AR**5**, the setting processing unit **114** may set or change the setting content and a sound source (reproduction content) by placing a higher priority on the use application of the specific area AR**2**. As described above, when the specific area AR**2** overlaps a part of the specific area AR**5**, the setting processing unit **114** may make the setting such that the setting content of the microphone-speaker device **2** located at the overlapping part reflects the setting content corresponding to the specific area AR**2**.
(89) Further, when the specific area AR**2** overlaps a part of the specific area AR**5**, the setting processing unit **114** may set a reproduction sound source of the microphone-speaker device **2** that is located inside the specific area AR**5** and outside the specific area AR**2** to a first sound source, and set a reproduction sound source of the microphone-speaker device **2** that is located inside the specific area AR**2** and at a part overlapping the specific area AR**5** to a second sound source.
(90) As yet another embodiment, the user may select a plurality of microphone-speaker devices **2** on a setting screen F**1**, and perform an operation to set the setting items collectively. For example, as illustrated in FIG. **13**, when the user selects respective images of the microphone-speaker devices **2**A and **2**D and changes the setting for mute to “OFF”, the setting processing unit **114** collectively sets the microphone-speaker devices **2**A and **2**D to “OFF” for mute.
(91) As described above, the setting processing unit **114** may receive, on an operation display **13**, an operation of selecting each of a plurality of microphone-speaker devices **2** from the user, group the plurality of microphone-speaker devices **2** selected by the user, and set the setting items for each of the grouped microphone-speaker devices **2**. For example, the setting processing unit **114** may set the reproduction language of the grouped microphone-speaker devices **2**, or assign weights to the speech of the users of the grouped microphone-speaker devices **2** in the minutes and set the level of importance to the speech.
(92) Further, the display processing unit **112** may color the frames surrounding images of the grouped microphone-speaker devices **2**, or vary the line type of the frames, for example, so that the grouped microphone-speaker devices **2** can be displayed to be distinguished from the other microphone-speaker devices **2** or other grouped microphone-speaker devices **2**. This allows the user to ascertain the grouped microphone-speaker devices **2** at a glance.
(93) Next, other methods of estimating the position of the microphone-speaker device **2** will be described. For example, each of the microphone-speaker devices **2** may be provided with a communication function such as a beacon, a tag, or the like. In this case, an estimation processing unit **111** can make signals of different frequencies be output from the external speakers **3***a* and **3***b* and estimate the position of each of the microphone-speaker devices **2** on the basis of a difference

in arrival time, i.e., the time when each of the microphone-speaker devices **2** has received the signals.

(94) Furthermore, as another example, when a plurality of microphones (microphone array), instead of the external speakers **3***a* and **3***b*, are installed in the meeting room R**1**, the estimation processing unit **111** can make a specific sound (a test sound, etc.) be reproduced from the microphone-speaker device **2**, and estimate the position of the aforementioned microphone-speaker device **2** on the basis of a difference in arrival time, i.e., the time when the microphone array has collected the specific sound. The microphone array is one example of the communication device of the present disclosure.

(95) Further, as yet another example, when a communication instrument (a receiver) such as a beacon or a tag is installed in the meeting room R**1**, the estimation processing unit **111** can make a signal be output from the microphone-speaker device **2**, and estimate the position of the microphone-speaker device **2** on the basis of a difference in arrival time, i.e., the time when the receiver has received the signal. The communication instrument (receiver) such as the beacon or tag is one example of the communication device of the present disclosure.

(96) In the embodiments described above, the controller **11** displays the setting screen F**1** on the operation display **13**, but as other embodiments, the controller **11** may display the setting screen F**1** on a device different from the control device **1** (i.e., a personal computer, a display, etc.). Further, the controller **11** may upload data on the setting screen F**1** to a cloud server, and the cloud server may allow the other devices to display the setting screen F**1**. For example, the cloud server may allow a smartphone of the user who attends the meeting to display the setting screen F**1**. By virtue of this feature, the user can ascertain the position and the setting content of each of the microphone-speaker devices **2** on his/her smartphone.

(97) The voice processing system of the present disclosure may be configured from the control device **1** alone or may be configured by combining the control device **1** with other servers such as a meeting server.

(98) Supplementary Notes of Disclosure

(99) An outline of the disclosure derived from the above embodiments will be described as supplementary notes. The configurations and the processing functions described in the following supplementary notes can be selected to be added or deleted and combined arbitrarily.

(100) Supplementary Note 1

(101) A voice processing system including a microphone-speaker device of a portable type carried by a user, and a communication device installed in a predetermined area in which the user is accommodated, the voice processing system comprising: an estimation processing unit which estimates a position of the microphone-speaker device relative to the communication device by executing transmission and reception of data between the microphone-speaker device and the communication device; and a display processing unit which displays, on a display, the position of the microphone-speaker device estimated by the estimation processing unit.

(102) Supplementary Note 2

(103) The voice processing system according to Supplementary Note 1, wherein the display processing unit displays, on the display, setting information of the microphone-speaker device.

(104) Supplementary Note 3

(105) The voice processing system according to Supplementary Note 2, further comprising a setting processing unit which receives, on the display, a setting operation for a setting item of the microphone-speaker device from the user, and sets the setting item according to the setting operation.

(106) Supplementary Note 4

(107) The voice processing system according to any one of Supplementary Notes 1 to 3, wherein: the voice processing system includes a plurality of microphone-speaker devices each corresponding to the microphone-speaker device; and the voice processing system further comprises a setting

processing unit which sets a specific place relative to the communication device to a first specific area, and varies setting content between a first microphone-speaker device located within the first specific area among the plurality of microphone-speaker devices and a second microphone-speaker device
(108) Supplementary Note 5
(109) The voice processing system according to Supplementary Note 4, wherein the setting processing unit sets a microphone and a speaker of the second microphone-speaker device to OFF.
(110) Supplementary Note 6
(111) The voice processing system according to Supplementary Note 4 or 5, wherein the setting processing unit sets a microphone gain of the first microphone-speaker device higher than a microphone gain of the second microphone-speaker device.
(112) Supplementary Note 7
(113) The voice processing system according to any one of Supplementary Notes 4 to 6, wherein the setting processing unit further sets a second specific area, which is a specific place relative to the communication device and is different from the first specific area, and sets each of the plurality of microphone-speaker devices to have setting content corresponding to the specific area where each of the microphone-speaker devices is located.
(114) Supplementary Note 8
(115) The voice processing system according to any one of Supplementary Notes 4 to 7, wherein when the second specific area overlaps a part of the first specific area, the setting processing unit makes setting such that setting content of the microphone-speaker device located at an overlapping part reflects setting content corresponding to the second specific area.
(116) Supplementary Note 9
(117) The voice processing system according to any one of Supplementary Notes 4 to 8, wherein when the second specific area overlaps a part of the first specific area, the setting processing unit sets a reproduction sound source of the microphone-speaker device that is located inside the first specific area and outside the second specific area to a first sound source, and sets a reproduction sound source of the microphone-speaker device that is located inside the second specific area and at a part overlapping the first specific area to a second sound source.
(118) Supplementary Note 10
(119) The voice processing system according to any one of Supplementary Notes 1 to 9, wherein: the voice processing system includes a plurality of microphone-speaker devices each corresponding to the microphone-speaker device; and the voice processing system further comprises a setting processing unit which receives, on the display, an operation of selecting the microphone-speaker device from the user, groups the plurality of microphone-speaker devices selected by the user, and sets a setting item for each of the grouped plurality of microphone-speaker devices.
(120) It is to be understood that the embodiments herein are illustrative and not restrictive, since the scope of the disclosure is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof are therefore intended to be embraced by the claims.

Claims

1. A voice processing system including a plurality of microphone-speaker devices, each of which is a portable type carried by a user, and a communication device installed in a predetermined area in which the user is accommodated, the voice processing system comprising: an estimation processing circuit which estimates a position of each of the plurality of microphone-speaker devices relative to the communication device by executing transmission and reception of data between each of the plurality of microphone-speaker devices and the communication device; a display processing circuit which displays, on a display, the position of each of the plurality of microphone-speaker

devices estimated by the estimation processing circuit; and a setting processing circuit which sets a specific place, relative to the communication device, to a first specific area, and makes different setting content between a first microphone-speaker device, among the plurality of microphone-speaker devices, located within the first specific area and a second microphone-speaker device, among the plurality of microphone-speaker devices, located outside the first specific area.

2. The voice processing system according to claim 1, wherein the display processing circuit further displays, on the display, setting information of at least one microphone-speaker device among the plurality of microphone-speaker devices.

3. The voice processing system according to claim 2, wherein the setting processing circuit receives, on the display, a setting operation for a setting item of the at least one microphone-speaker device from the user, and sets the setting item according to the setting operation.

4. The voice processing system according to claim 2, wherein the setting processing circuit: receives, on the display, an operation of selecting a set of microphone-speaker devices, among the plurality of microphone-speaker devices, from the user, groups the set of microphone-speaker devices selected by the user, and sets a setting item for each of the grouped set of microphone-speaker devices.

5. The voice processing system according to claim 1, wherein the setting processing circuit sets a microphone and a speaker of the second microphone-speaker device to OFF.

6. The voice processing system according to claim 1, wherein the setting processing circuit sets a microphone gain of the first microphone-speaker device higher than a microphone gain of the second microphone-speaker device.

7. The voice processing system according to claim 1, wherein the setting processing circuit further sets a second specific area, which is a specific place relative to the communication device and is different from the first specific area, and sets each of the plurality of microphone-speaker devices to have setting content corresponding to a specific area where each of the plurality of microphone-speaker devices is located.

8. The voice processing system according to claim 7, wherein, when the second specific area overlaps a part of the first specific area, the setting processing circuit makes a setting such that setting content of a microphone-speaker device, among the plurality of microphone-speaker devices, located at an overlapping part reflects setting content corresponding to the second specific area.

9. The voice processing system according to claim 7, wherein, when the second specific area overlaps a part of the first specific area, the setting processing circuit sets a reproduction sound source of a microphone-speaker device, among the plurality of microphone-speaker devices, that is located inside the first specific area and outside the second specific area to a first sound source, and sets a reproduction sound source of another microphone-speaker device, among the plurality of microphone-speaker devices, that is located inside the second specific area and at a part overlapping the first specific area to a second sound source.

10. A voice processing method which involves use of a plurality of microphone-speaker devices, each of which is a portable type carried by a user, and a communication device installed in a predetermined area in which the user is accommodated, the voice processing method being executed by one or more processors and comprising: estimating a position of each of the plurality of microphone-speaker devices relative to the communication device by performing transmission and reception of data between each of the plurality of microphone-speaker devices and the communication device; displaying the estimated position of each of the plurality of microphone-speaker devices on a display; setting a specific place, relative to the communication device, to a first specific area; and making different setting content between a first microphone-speaker device, among the plurality of microphone-speaker devices, located within the first specific area and a second microphone-speaker device, among the plurality of microphone-speaker devices, located outside the first specific area.

11. A non-transitory computer-readable recording medium having recorded thereon a voice processing program comprising instructions for a plurality of microphone-speaker devices, each of which is a portable type carried by a user, and a communication device installed in a predetermined area in which the user is accommodated, the voice processing program causing one or more processors to execute: estimating a position of each of the plurality of microphone-speaker devices relative to the communication device by performing transmission and reception of data between each of the plurality of microphone-speaker devices and the communication device; displaying the estimated position of each of the plurality of microphone-speaker devices on a display; setting a specific place, relative to the communication device, to a first specific area; and making different setting content between a first microphone-speaker device, among the plurality of microphone-speaker devices, located within the first specific area and a second microphone-speaker device, among the plurality of microphone-speaker devices, located outside the first specific area.